Fiscal Year:	FY 2008	**Task Last Updated:**	FY 10/08/2008
PI Name:	Booksh, Karl Steven Ph.D.		
Project Title:	Development of a Modular, Fiber Optic Surface Plasmon Resonance Sensor for Quantitation of Diagnostic Proteins for Healing of Burns and Wounds		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
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Comments:			
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
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Contact Monitor:		**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 4/30/2008 per K. Major/NSBRI (3/08)		
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Task Description:	We propose to develop a fiber optic sensor system capable of in vitro and ultimately in vivo quantifying clinically relevant levels of diagnostic proteins in wounds including burns. The sensor system is designed for rapid field portable analysis without need for laboratory support; by design all reagents for sensor preparation and sample analysis will be contained in a small 10 ml to 25 ml vial. The sensor system is modular to allow expansion to multiple analytes and applications beyond wound healing. By constructing the sensors on optical fibers, the sensor system can employ inexpensive, interchangeable sensing elements for specific biological tests with simple automated procedures for preparing the sensing elements for operation. This provides flexibility and timely responsiveness to address multiple health related issues where protein biomarkers have diagnostic value. Also, the fiber optic based sensing elements may be compatible with future in vivo analyses, and the in vitro development work proposed here will be highly relevant when the in vivo applications are pursued. The sensors use surface plasmon resonance (SPR) on optical fibers to detect the diagnostic biomarkers through immunoassays. The fibers are encapsulated in a thin film polymer housing that prevents cellular fouling of the sensing region. Because the assay is conducted on the surface of an optical fiber, the probes can be used as immersion probes in vitro in drawn fluid or as minimally invasive in vivo sensors that can be placed directly on a burn or open wound (future work). To meet these specific aims, we can demonstrate our capability to construct polymer housings with pore sizes larger than proteins but smaller than cells to protect and shield the sensors surface (S1). We have evidence that judicious choice and functionalization of biopolymer scaffolds can simultaneously enhance the sensor sensitivity to target analytes (S2.1) while significantly reducing the degree of protein fouling on the sensor (S2.2) relative to SPR standard functionalized dextran hydrogel scaffolds. Preliminary tests in saline solution have demonstrated that the sensor system achieves physiologically relevant detection limits for 2 of the target biomarkers: tumor necrosis factor alpha and interleukin 6(S3).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Working with RSL we are developing the sensor as a portable diagnostic tool to for strokes and heart attacks, and wound healing. The idea is that the sensor system and per-analysis cost will be inexpensive such that clinics, rural hospitals, and first responders can more rapidly and reliably determine the existence and severity of heart attacks and strokes. Ideally we will be able to differentiate between ischemic and hemorrhagic events in time to apply appropriate pharmaceutical countermeasures (3 hours post incident). For wound healing we are aiming at a home test kit for diabetic ulcers and bedsores.
Task Progress:	Improving reliability of surface coatings. The quantitative performance of the organic coatings that we have employed has varied from batch to batch of the thiol-linking layer that we synthesize to attach antibodies to the gold surface. We are looking into the variables that are effecting the quantitative performance of the sensors. The observed variance is in the rate of mass transport to the sensor surface, not the total saturation signal observed. We believe that an impurity in our attachment chemistry is improving the rate by drawing the proteins to the surface via electrostatic attraction. We are doing final experiments to validate the impurity hypothesis. At the same time, we are investigating the identity of impurities in the organic linker as well as adding additives to the self assembled monolayers that will facilitate transport of proteins in order to improve reaction kinetics. Eliminating dependence on antibodies. We are observing a 2 to 4 fold variation in antibody binding constants between vendors for one target application and up to a two fold variation in antibody binding constants between lots within one vendor. Also, it is common to lose an order of magnitude of binding coefficient from tethering an antibody to the sensing surface. As mentioned above, we are working to combine SPR with electrophoretic separation methods where the isoelectric focusing will provide up to 10,000 fold preconcentration of the analyte on the sensing pads. This will minimize the need for antibody receptors for target proteins and open the door to detection of smaller molecules. Demonstrating SPR on a microfluidic chip. The key technological advancement required to combine SPR with microfluidics based electrophoretic separations is to monitor the SPR signal from small, 1 micron by 1 micron, sensing pads embedded in the separation channel. Employing a continuous gold film is unacceptable because embedding a conducting layer in the microfluidic channel would short-circuit the electrophoretic separation; plus the propagation length of a freely propagating SPR signal on a continuous gold surface is approximately 40 microns. In the electrophoretic channels, the pads must be spaced at least 500 nm apart or localized SPR on each sensing area begins to interact with neighboring sensing pads. We have successfully tested the ability to image arrays of small closely packed sensing pads – close enough together to realize micron-scale spatial resolution but far enough apart to eliminate interactions.
Bibliography Type:	Description: (Last Updated:)